REPORT · 2026-11-02

Alberto AI v1.8.4 — Real Provider Benchmark

10/10 passed - nemotron-3-ultra

AUTHOR

DATE

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Executed with `scripts/benchmark_alberto.py` (real provider calls, **no mocks**). Script came from `audit/real-benchmark` branch.

Configuration

Item	Value
Agent version	1.0.0
Primary model	`nvidia/nemotron-3-ultra-550b-a55b`
Fallback	`nvidia/nemotron-3.5-lightning-30b-a3b` (thinking enabled)
Tasks configured	chat, code, embed, reasoning, summary, vision
Sandbox	LocalSandbox (host-process)
Repeats	1 per case

Results — 10/10 passed (100%)

Metric	Value
Cases	10
Passed	**10 (100%)**
Errors	0
Mean latency	13,477 ms
Median latency	13,096 ms
p95 latency	16,590 ms

By category

Category	Cases	Pass rate	Median latency
arithmetic	2	100%	14,677 ms
classification_pt	1	100%	13,785 ms
instruction_following	3	100%	12,439 ms
language	1	100%	13,386 ms
multilingual	1	100%	12,938 ms
reasoning	1	100%	14,606 ms
structured_output	1	100%	12,890 ms

All cases answered in pt-BR and en as required, including strict-format (JSON-only, exact-string) and multi-step arithmetic.

Latency context (honest comparison)

The 13.1 s median reflects **nemotron-3-ultra-550b with reasoning enabled** — a 550B model doing step-by-step thinking. Published framework benchmarks use much smaller non-thinking models:

Framework (agentmail 90-test suite)	Median
Mastra	2.2 s
Pydantic AI	2.2 s
LangGraph	2.5 s
OpenAI Agents SDK	2.5 s
AG2	2.8 s
Google ADK	3.4 s
CrewAI	4.0 s
Claude Agent SDK	8.5 s
Alberto (nemotron-3-ultra, thinking)	**13.1 s**

To compare apples-to-apples, re-run with a small non-thinking model (e.g. `meta/llama-3.1-8b-instruct`) — that isolates framework overhead from model latency.

Reproduce

```
export NVIDIA_API_KEY=...
export NVIDIA_API_KEY_FALLBACK=...
alberto model set chat nvidia nvidia/nemotron-3-ultra-550b-a55b \
  https://integrate.api.nvidia.com/v1 NVIDIA_API_KEY
python scripts/benchmark_alberto.py --repeats 3 --pause 2
```

Raw result: `benchmark-results/latest-real-benchmark.json`

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